stat inference

hpye

«known»

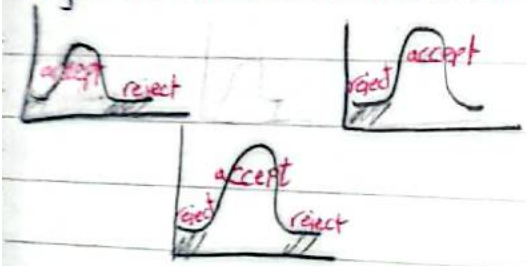
1000

reject — accept

$H_0: \mu = value$ $\begin{cases} + \\ - \\ 0 \end{cases}$

H_1

$H_1: \mu > value$ (Right tail) — $H_1: \mu \neq value$ (two tail) — $H_1: \mu < value$ (left tail)



point estimate

«unknown»

- $\bar{X}$

Confidence interval

Price

lower upper: 7000 . 15000 — 95% — 5%

lower upper: 8000 . 10000 — 90% — 10%

5000 estimat

«unknown»

parameters:

- mean (μ) — متوسط المجتمع
- variance (σ^2) — تباين المجتمع
- standar deviation (σ) — انحراف معياري
- propotion (θ) — النسبة

Stat:

- $\bar{X}$ — متوسط العينة
- S^2 — تباين العينة
- S — انحراف معياري للعينة

$[\bar{X} \pm \text{estimate error}]$

lower: $\bar{X} - \text{estimate error}$

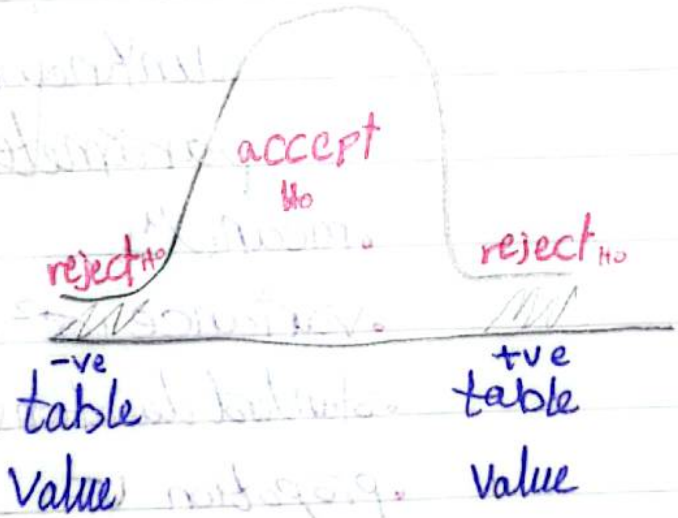
upper: $\bar{X} + \text{estimate error}$.

$\left[\bar{X} \pm \text{tabled value} \times \frac{\text{standar}}{\sqrt{\text{sample size}}}\right]$

lower: $\bar{X} - \text{tabled value} * \frac{\text{standar deviation}}{\sqrt{n}}$

upper: $\bar{X} + \text{tabled value} * \frac{\text{standar deviation}}{\sqrt{n}}$

$P[\text{lower} < \mu < \text{upper}] = 90\%, 95\%, 99\%$

* test stat = $\dfrac{\bar{X} - \mu_0}{\dfrac{\text{standar deviation}}{\sqrt{\text{sample size}}}}$

* table value =